TripNest AI: AI and ML Documentation

Version 1.0 · 2 November 2026. Everything below describes what is implemented. Numbers are read from `backend/models/metrics.json` (trained 1 November 2026).

1. What is machine learning here, and what is not

Part	Method	Is it ML?
Crowd prediction	Gradient-boosted regression trees (scikit-learn `HistGradientBoostingRegressor`) trained on hourly history	**Yes**
Recommendation	Transparent weighted score of the model's predictions, the interest match, distance and season	**No.** It is a rule-based score that *consumes* ML predictions. We do not call it AI
Crowd levels	Fixed thresholds on predicted utilisation	No
City search	Static list with text matching	No

2. Data

Demonstration history. There is no verified public source of hourly visitor counts for Indian attractions, so `services/crowd_sim.py` generates 365 days of hourly counts for each of 271 attractions from an assumed carrying capacity and popularity, the daily rhythm of the attraction category, weekday, season, national holidays and simulated rain, plus noise. **It is not real data.** To use real data, supply a table with the columns `key, date, hour, count, rain` and retrain.

Target. Utilisation = people present ÷ assumed capacity, modelled as `log(utilisation + 0.02)`. This is scale-free, so one model serves every attraction and a new attraction gets forecasts at once (no attraction ID is used as a feature).

Features (15). category, climate, hour, weekday, weekend, public holiday, festival, month, in-season, rain, popularity, log capacity, fraction of opening hours elapsed, covered, ecologically sensitive.

3. Training and evaluation (reproducible)

Run `cd backend && python -m scripts.train_model --days 365` (about 40 s).

Split	Dates	Rows
Train	2025-11-02 to 2026-10-04	1,008,641
Validation	2026-10-05 to 2026-10-18	41,902
Test	2026-10-19 to 2026-11-01	41,902

Chronological split (no shuffling). Two candidate settings were compared on the validation fortnight and the better one (learning rate 0.10, 63 leaves, 250 iterations) was refit on train plus validation and scored **once** on the test fortnight. The final serving model is trained on all history.

Test fortnight (hourly, 41,902 rows)	MAPE	WAPE	MAE (people)	R^2
TripNest model	**9.38%**	8.64%	54.8	0.986
Baseline: same hour one week earlier	31.79%	28.51%	181.1	0.730
Baseline: mean by attraction, weekday and hour	27.45%	29.89%	189.8	0.774

MAPE ignores hours with fewer than 10 people. **These results are measured on simulated data generated by the same kind of rules the model learns, so they show the pipeline works and beats**

simple baselines; they say nothing about accuracy on real crowds. Strongest inputs by permutation importance: category, hour, popularity, weekday, holiday, opening-hours fraction.

4. Uncertainty

The standard deviation of the model's log-errors on the test fortnight gives an 80% likely range (±1.28σ) per attraction, shown as the band on charts and as "likely low to high". A destination's range adds its attractions' ranges, so it is wider than a true destination-level interval. The chance of reaching capacity is available per attraction forecast (`p_exceed`).

5. Levels

Low below 40% of assumed capacity, Moderate 40–70%, High 70–90%, At Capacity above 90%. Ecologically sensitive attractions use 25%, 50% and 70%. A destination's level uses its attractions' total people ÷ total capacity for the hour.

6. Recommendation score

`score = Σ weight × factor` over the factors that apply: **crowd 0.40** (1 − busiest daytime utilisation ÷ 0.9, floored at 0, capped at 1), **interest 0.25** (1 if a destination category matches, 0.5 if only an attraction matches, else 0), **distance 0.20** (1 − km ÷ 2500, straight-line), **season 0.15** (1 in a best month, else 0.3). Without an interest or a start, that factor's weight is 0 and the rest are re-normalised to sum to 1. **Popularity is not used.** The score is shown, with each factor and a plain-language reason.

7. Provenance and honesty

Every prediction response and screen says: ML prediction, demonstration data, not live, when it was made, when the model was trained, how fresh the history is, how uncertainty is measured, and four limitations. No live counts, ticketing or sensor data are used or implied. Retraining on real data is supported but has **not** been done.

8. Limitations

Simulated history; assumed capacities; no knowledge of local events or weather on the day beyond seasonal rain probability; straight-line distance; the 77-city list; evaluation only on the generator's own patterns; no fairness, drift or monitoring tooling.